INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 8: Appendix

by

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8 Appendix

8.1 Evaluation Pipeline and Reproducibility

This appendix documents the evaluation pipeline behind Chapter 4 and behind the registered analysis of Appendix 8.6: its stages, the extraction design, the account-type classification, the timed runs and the default parameters.

8.1.1 Pipeline stages

The evaluation source is in the folder `1-Dissertation-Works/ERC20-Allowance-PageRank-Wallet-Reputation/experiments` of the public repository https://github.com/abitocodes/phd_works, and the paths below are relative to it. The evaluation has five stages. The first three run on Arbitrum One over the observation window, 1 October 2023 to 30 June 2026 (UTC, inclusive at both ends); the fourth adds July to September 2026, and the fifth reads only the tables that the first four left in BigQuery.

1. Plan, cohort and graphs: Commit the analysis plan (`docs/analysis_plan.md`) and the configuration before any log of the study is extracted (commit `43d8045`, 13:41 UTC on 9 October 2026). Copy the logs of the window into a project table (scan A), list the spenders that at least three owners approved (`sql/02_spender_candidates.sql`), read their account types (`scripts/check_account_code.py`) and keep the contracts as the plan's cohort (`scripts/build_cohort.py`). Extract the ego events of the cohort (`sql/03_ego_events.sql`) and aggregate them into the pair tables of both graphs at T_{obs} and at t_1 (`sql/04_anchor_pairs.sql`).
2. Checks and alignment: Score every method at T_{obs} with the solver settings of Section 3.1, compute the checks of Section 3.9 and report Spearman ρ and Kendall τ with 95% paired-

bootstrap intervals (400 resamples, seed 42) and the contrasts of Section 3.10 (`scripts/run_same_window.py`).

3. Temporal holdout W0: Score every method at t_1 and evaluate the spender and trader cohorts against the labels of April–June 2026, with the contrasts of rule A (`scripts/run_holdout.py`).
4. Registered window W1: Commit the registration (`docs/registration_w1.md`) with the status in the configuration set to registered (commit `0d3a057`, 16:49 UTC on 9 October 2026). `scripts/extract_w1.py`, which will not run before that commit, records the hashes of both files in an extraction manifest and runs scan B and the ego extraction of July–September; `scripts/run_registered.py` evaluates rule B and the comparison on outcomes built from neither edge type.
5. Revision: Commit the revision of the amounts before any revised score (commits `a897475`, `09dfc32` and `2705b28` on 10 October 2026; Section 3.5.1). `scripts/usd_inputs.py` builds the token list and the daily prices, `scripts/chainlink_check.py` compares them with the Chainlink feeds, and `scripts/run_usd_pipeline.py` runs `sql/12` to `sql/19`, which cut the revised cohort, pair tables, proxies and labels from the ego tables of stages 1 and 4. With `CONTRACT_REP_VARIANT=usd` the scripts of stages 2 to 4 then compute the revised analysis, and `scripts/run_robustness.py`, `scripts/sybil_model.py` and `scripts/run_benchmark.py` add the damping, value-slope and sample-definition checks, the Sybil model and the timed runs.

The order of these steps is recorded in the commit history, in the cost ledger and in the extraction manifest of W1. The digest of the revised summaries records the code revision `2705b28` and names five files that differed from it; they were committed with the revised summaries in `77e71f5`. The timed runs were made on an otherwise idle machine with an Intel Core Ultra 9 185H processor and 64 GB of memory under Windows 11 (build 26200), with Python 3.11.9, NumPy 2.3.2 and SciPy 1.16.1. `scripts/export_latex.py` wrote the tables to `results/tables/` and the figures to `Fi`

gures/generated/ from the summaries.

The experiments folder holds the queries, scripts and tests, the versioned configuration (config/contract_reputation.yaml), the plan, the registration and the revision document (docs/), the token list and daily prices with their Chainlink comparison (data/0-usd-token-list-and-prices/) and the cost ledger. The summaries behind the tables are in data/3-published-results-for-thesis/revised-usd/, and those of the registered analysis in data/3-published-results-for-thesis/. Each of the two folders holds copies of the configuration, the plan and the registration, a digest results_digest.md, which names the summary file and key of every number, and SHA256SUMS.txt, so any number in a table can be checked against its summary without rerunning the pipeline. The extracted tables and the per-contract scores, stored as parquet parts below 90 MB, are kept in a public Google Drive folder; the committed file drive-manifest.json gives its link and lists every part with its checksum.

Synthetic tests check the solver, the edge weights, the paired bootstrap and the US-dollar conversion offline. Real extraction needs Google Cloud credentials and a billing project.

8.1.2 BigQuery extraction design

The public table bigquery-public-data.goog_blockchain_arbitrum_one_us.logs is read once for each window. Scan A copies the ERC-20 Approval and Transfer logs of the observation window, the GMX V2 position closes and the Aave V3 pool events, which no analysis of this thesis uses, into one project table partitioned by month, and scan B does the same for July to September 2026. Every later query reads these project tables, so the chain-wide scan is paid for once per window, and the ego extraction keeps only events with a cohort contract at one end, which ties the size of the extract to the cohort instead of the whole chain. Appendix 8.2 shows the queries.

Every job is listed in data/bq_ledger.json with its query, parameters, estimated and billed bytes and cost. The ledger holds 32 jobs that billed 5.57 TiB, or USD 34.80 at the on-demand rate of USD 6.25 per TiB, against the budget of USD 50 fixed in the configuration. Scan A alone billed 3.29 TiB (USD 20.55) and scan B 0.26 TiB (USD 1.64); the 17 jobs of the registered analysis billed

USD 29.71 and the 15 jobs of the revision USD 5.09. Three jobs of the revision ran outside `scripts/run_bq.py` and were entered in the ledger by hand.

8.1.3 Account-type classification

Every address that could enter a cohort was classified by its code under the rule of Section 3.2. `scripts/check_account_code.py` calls `eth_getCode` for each address at one block, which it records with the time of the read. Table 8.1 lists the four runs, all made on 9 October 2026 for the address sets of the plan, in which approvals of every token counted. The first $20{,}000$ candidates were read from the public endpoint https://arb1.arbitrum.io/rpc, which then limited the request rate, and the other $9{,}022$, at the same block, from https://arbitrum-one.public.blastapi.io, which also served the later runs.

Table 8.1: Account types read with `eth_getCode` on Arbitrum One, one block per run. Contract: non-empty code that is not a delegation designator. No code and EIP-7702 (a delegation designator) are both externally owned accounts. The address sets are those of the plan (approvals of any token); the cohorts of the revised analysis are subsets of them. The W0 run read the spenders that were not cohort candidates, and the W1 run the addresses that no earlier run had classified. Sources: the JSON files beside the outputs in `data/1-raw-blockchain-extracts/cohort/` and `gmx/`.

Address set	Block	Read at (UTC)	Addresses	Contract	No code	EIP-7702
Cohort candidates: non-zero approvals from ≥ 3 owners, observation window	513,216,506	9 Oct 2026, 13:55	29,022	27,844	1,121	57
GMX V2 accounts with ≥ 3 closes in the observation window	513,219,999	9 Oct 2026, 14:10	40,307	7,633	28,075	4,599
W0 spenders with a positive allowance at t_1, not cohort candidates	513,233,527	9 Oct 2026, 15:08	8,224	7,792	413	19
W1 spenders and GMX V2 accounts with ≥ 3 closes in W1, not read before	513,257,476	9 Oct 2026, 16:52	1,153	350	568	235

The matched cohort is part of the plan's cohort and was classified in the first run. The W0 spenders were classified in the first and third runs, since a spender with a positive latest allowance of one of the five tokens at t_1 also held that allowance in the ego data of the plan. The W1 spenders and the trader cohorts take their account types from all four runs, so the revision made no new read. Every read refers to the chain on 9 October 2026, and an address is classified by the code it held on that day.

8.1.4 Benchmark run logs

Table 8.2 lists the wall-clock seconds of every timed solver call of the benchmark, in the order the runs were made, with their mean and standard deviation. Each configuration had one untimed warm-up before its five timed runs, and each graph and setting was timed once: the $d = 0.85$ rows of Table 4.6 and the full-cohort row of Table 4.2 repeat the first block of this table.

Table 8.2: Wall-clock seconds of every timed solver call of the benchmark (matched cohort, $n = 15{,}107$ contracts), in the order the runs were made, with their mean and standard deviation. Each configuration had one untimed warm-up before its timed runs, and each graph and setting was timed once: the $d = 0.85$ rows of Table 4.6 and the full-cohort row of Table 4.2 are the first block, reused. The stages are the subsamples of Table 4.2.

Method	Run 1	Run 2	Run 3	Run 4	Run 5	Mean	SD
Full graphs, $d = 0.85$							
EndorseRank	9.570	11.063	10.497	9.811	10.530	10.294	0.601
AWP	21.139	20.446	24.122	24.492	21.637	22.367	1.825
C-PR ($\lambda = 0.5$)	33.813	39.576	35.439	35.793	36.187	36.162	2.112
S-PR	38.769	36.970	29.310	35.214	31.848	34.422	3.833
Full graphs, $d = 0.75$							
EndorseRank	6.045	5.963	6.003	5.984	5.927	5.984	0.044
AWP	13.208	16.968	16.667	15.870	13.469	15.236	1.781
Full graphs, $d = 0.95$							
EndorseRank	31.701	32.171	30.886	30.574	30.836	31.233	0.673
AWP	58.498	56.896	59.713	64.490	65.212	60.962	3.697
Stage $n = 1{,}000$							
EndorseRank	0.425	0.489	0.585	0.683	0.719	0.580	0.125
AWP	1.892	1.408	1.870	1.758	1.458	1.677	0.229
Stage $n = 2{,}000$							
EndorseRank	0.944	0.956	0.925	0.932	0.937	0.939	0.012
AWP	3.748	4.876	4.133	4.811	3.792	4.272	0.543
Stage $n = 5{,}000$							
EndorseRank	2.293	2.420	2.489	2.555	2.723	2.496	0.159
AWP	8.713	9.861	9.895	8.051	8.453	8.995	0.840
Stage $n = 10{,}000$							
EndorseRank	6.253	6.375	6.065	6.285	6.119	6.220	0.126
AWP	17.640	17.362	17.160	16.124	15.816	16.821	0.802

8.1.5 Parameter summary

Table 8.3 collects the default parameters of the revised analysis; every value is read from the versioned configuration or, for the tokens, prices, value slopes and cap, from the inputs committed with the revision (Section 3.5.1).

Table 8.3: Default parameters used in the evaluation.

Parameter	Value	Role
PageRank damping d	0.85; sweep 0.75, 0.95	Teleport mixture
Convergence tolerance ε	10^{-8} in L_1 norm	Power-iteration stop
Max iterations $I_{\max}$	300	Power-iteration cap
Logistic k	0.01	Decay slope (AWP, EndorseRank, transfer layer of C-PR)
Logistic t_0	180 days	Decay midpoint (AWP, EndorseRank, transfer layer of C-PR)
Value transform b	$\ln 3/m$ per dollar, $m_T = 200$, $m_A = 62$	Value weight $V(z)$ of AWP (m_T) and EndorseRank (m_A)
Allowance cap U	20,000 dollars	C-PR allowance layer; in-approve value
Tokens and prices	WETH, USDC, USDT0, ARB, WBTC; DefiLlama daily close	Events and US-dollar amounts (Table 8.4)
Restart distribution	$\propto X_u$; uniform	AWP; EndorseRank, C-PR
Coupling weight λ	0.5 primary; 0.25, 0.75	C-PR allowance-layer weight, fixed in the plan
S-PR restart vector	C-PR ($\lambda = 1$) on the transfer-layer nodes, renormalized	Endorsement-seeded teleportation
Observation window	2023-10-01 – 2026-06-30	UTC inclusive; T_{obs} anchors the decay
W0 freeze / labels	2026-03-31 / 2026-04-01 – 2026-06-30	Spender and trader cohorts; rule A
W1 freeze / labels	2026-06-30 / 2026-07-01 – 2026-09-30	Spender and trader cohorts; rule B, margin $\delta = 0.01$
Bootstrap resamples / seed	400 / 42; 2,000 / 42	Paired contract bootstrap; trader comparisons
Timed runs / warm-up	5 / 1	Runtime protocol
Scaling seed	`contract-benchmark-v1`	Deterministic SHA256 subsamples
BigQuery budget	USD 50	Checked before every query

8.2 SQL Templates

This appendix shows the three query patterns on which the graphs of both analyses rest, abridged; the full files are in `sql/` of the experiments folder (Appendix 8.1.1). Placeholders in double underscores, such as `__DEST__`, are filled in by `scripts/run_bq.py` before a query runs, and names that start with `@` are query parameters.

8.2.1 Log materialization pattern

Scan A and scan B run the same query, `sql/01_materialize_logs.sql`, over different windows. It reads the public log table once, tags each log by kind and keeps the four kinds the study needs in a project table partitioned by month. With the topic hashes and the GMX and Aave branches shortened, it reads:

```
CREATE TABLE `__DEST__`
PARTITION BY TIMESTAMP_TRUNC(block_timestamp, MONTH) CLUSTER BY kind AS
WITH tagged AS (
  SELECT *, CASE
      WHEN ARRAY_LENGTH(topics) = 3 AND topics[OFFSET(0)] = '0x8c5b...b925'
        THEN 'A'                                  -- ERC-20 Approval
      WHEN ARRAY_LENGTH(topics) = 3 AND topics[OFFSET(0)] = '0xddf2...b3ef'
        THEN 'T'                                  -- ERC-20 Transfer
      WHEN ... THEN 'G'                           -- GMX V2 PositionDecrease
      WHEN ... THEN 'V'                           -- Aave V3 pool events
    END AS kind
  FROM `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`
  WHERE block_timestamp >= TIMESTAMP(@start_ts)
    AND block_timestamp < TIMESTAMP(@end_ts))
SELECT block_timestamp, block_number, log_index, kind,
  FROM_HEX(SUBSTR(address, 3)) AS emitter,
  IF(kind IN ('A', 'T'), FROM_HEX(SUBSTR(topics[OFFSET(1)], 27)), NULL) AS a,
  IF(kind IN ('A', 'T'), FROM_HEX(SUBSTR(topics[OFFSET(2)], 27)), NULL) AS b,
  IF(kind IN ('A', 'T'), SAFE.FROM_HEX(SUBSTR(data, 3)), NULL) AS val -- ...
FROM tagged WHERE kind IS NOT NULL
```

For an approval, `a` is the owner and `b` the spender; for a transfer, `a` is the sender and `b` the recipient. In both, `emitter` is the token contract and `val` the 32-byte amount, and addresses are stored as 20-byte values.

8.2.2 Ego-network extraction pattern

`sql/03_ego_events.sql` reads the materialized table of one window and keeps the approvals and transfers with a contract of the plan's cohort, held in `contract_rep.cohort`, at one end. Abridged, for the approvals:

```
CREATE OR REPLACE TABLE `dissertation-bq.contract_rep.approvals___SUFFIX__`
PARTITION BY TIMESTAMP_TRUNC(block_timestamp, MONTH) AS
SELECT l.block_timestamp, l.block_number, l.log_index,
       l.emitter AS token_address, l.a AS owner, l.b AS spender,
       u256(l.val) AS value
FROM `dissertation-bq.contract_rep.__SRC__` AS l
WHERE l.kind = 'A'
  AND l.block_timestamp >= TIMESTAMP(@start_ts)
  AND l.block_timestamp < TIMESTAMP(@end_ts)
  AND (l.a IN (SELECT address FROM `dissertation-bq.contract_rep.cohort`)
```

```
       OR l.b IN (SELECT address FROM `dissertation-bq.contract_rep.cohort`));
-- transfers___SUFFIX__: the same with kind = 'T', from_address = a,
-- to_address = b and zero amounts dropped: AND LTRIM(TO_HEX(l.val), '0') != ''
```

The function u256, defined at the top of the file, reads the 32-byte amount as a double in raw base units. Approvals of zero are kept, because an approval of zero sets an allowance to zero, while zero-value transfers are dropped. The revised analysis does not run this query again: `sql/12_usd_ego.sql` forms the revised cohort from these tables, keeps the events of the five tokens with one of its contracts at one end and values a transfer at its token's price on the UTC day of its block.

8.2.3 Latest-allowance and pair-table pattern

In the revised analysis `sql/14_usd_anchor_pairs.sql` turns the ego events up to one anchor into the edge lists of both graphs; `sql/04_anchor_pairs.sql` did the same for the registered analysis in raw base units, with $b = 1$ and no cap. Abridged:

```
CREATE OR REPLACE TABLE `dissertation-bq.contract_rep.u_allow_triples___TAG__` AS
SELECT r.token_address, r.owner, r.spender, r.amount * p.price_usd AS usd,
       TIMESTAMP_DIFF(TIMESTAMP(@anchor_ts), r.block_timestamp, MICROSECOND)
         / 8.64e10 AS age_days
FROM (
  SELECT *, ROW_NUMBER() OVER (
            PARTITION BY token_address, owner, spender
            ORDER BY block_number DESC, log_index DESC) AS rn
  FROM `dissertation-bq.contract_rep.u_approvals`
  WHERE block_timestamp >= TIMESTAMP(@start_ts)
    AND block_timestamp <= TIMESTAMP(@anchor_ts)
) AS r
JOIN `dissertation-bq.contract_rep.u_prices` AS p
  ON p.token = r.token_address AND p.day = DATE(TIMESTAMP(@anchor_ts))
WHERE r.rn = 1 AND r.value > 0;

CREATE OR REPLACE TABLE `dissertation-bq.contract_rep.u_allow_pairs___TAG__` AS
SELECT owner, spender,
       SUM(sig(age_days) * vt(usd, @b_allowance)) AS w_er,  -- EndorseRank
       SUM(LEAST(usd, @cap_allowance)) AS w_raw,    -- C-PR allowance layer
       MAX(sig(age_days)) AS max_sig
FROM `dissertation-bq.contract_rep.u_allow_triples___TAG__`
GROUP BY owner, spender;
```

```
-- u_transfer_pairs___TAG__: per (from_address, to_address) up to the anchor,
-- w_awp = SUM(sig * vt(usd, @b_transfer)), w_raw = SUM(usd * sig), MAX(sig)
```

The functions `sig` and `vt`, defined at the top of the file, are the logistic decay of Equation (3.3) with $k = 0.01$ and $t_0 = 180$ days and the value transform of Equation (3.4), set to one from $bz = 40$ on, where it equals one in double precision. The window function picks the latest approval of each token, owner and spender by block number and log index; it is kept only when its value is positive, so a revocation removes that token from the pair, and it is valued at the price of the anchor's day. The slopes $b_A = \ln 3/62$ and $b_T = \ln 3/200$ per dollar and the cap $U = 20{,}000$ dollars are passed as parameters from `constants.json`. Each pair row carries the weights of every method, so all scores at one anchor come from the same rows. The same query also writes the EndorseRank and AWP weights with the slope multiplied and divided by ten, from which the value-slope check of Table 4.7 is computed.

8.3 Elementary Properties of PageRank and Rank Correlation

Chapters 2–4 rely on the elementary properties summarized here. The material is standard: the PageRank properties follow Page et al. (1999) and Langville and Meyer (2006), and the rank correlations Spearman (1904) and Kendall (1938).

8.3.1 Existence and uniqueness of PageRank

Take P to be a row-stochastic matrix on $|V|$ nodes, possibly after the dangling-node repair of Equation (2.1). For $d \in (0, 1)$ the Google matrix of Equation (2.2),

$$G_{\mathrm{PR}} = dP^{\top} + \frac{1-d}{|V|}\mathbf{1}\mathbf{1}^{\top},$$

is positive and column-stochastic. By the Perron–Frobenius theorem for positive matrices, G_{PR} has a single eigenvalue of modulus one, and the corresponding eigenvector $\mathbf{r}$ is strictly positive; we can scale it so that $\sum_i r_i = 1$. It follows that, once the damping and the dangling-node rules are fixed,

every directed graph has unique PageRank scores (Langville & Meyer, 2006).

Two restart vectors in this thesis are not uniform. AWP's, proportional to activeness, is zero at every address that has sent nothing, and S-PR's, the normalized score of the allowance walk, is zero at every node of the transfer layer that has no allowance score, so neither Google matrix is positive. The fixed point is still unique: for two probability vectors the restart term cancels in their difference, and the operator multiplies their $\|\cdot\|_1$ distance by at most d. In general it is not strictly positive, however, because a node with no restart mass that no path from a restart node reaches keeps a score of zero. Uniqueness means that, once its parameters are fixed, each method assigns exactly one score vector to a given edge set, so any difference between $\mathbf{s}_{\text{ER}}$ and $\mathbf{s}_{\text{AWP}}$ arises from their edges and restart distributions.

8.3.2 Continuity in edge weights

For fixed $d \in (0, 1)$, PageRank depends continuously on the entries of P. If $\tilde{P}$ is a second row-stochastic matrix on the same nodes, with PageRank vector $\tilde{\mathbf{r}}$, then

$$\|\mathbf{r} - \tilde{\mathbf{r}}\|_1 \leq \frac{d}{1-d} \max_i \left\| P_{i\cdot} - \tilde{P}_{i\cdot} \right\|_1,$$

a bound that does not depend on $|V|$ (Langville & Meyer, 2006).

Continuity in P is not the same as continuity in the edge weights. A row $P_{i\cdot} = w_{i\cdot}/o_i$ moves continuously with the weights only while the out-strength o_i stays positive. When a node gains its first out-edge or loses its last one, its row jumps between the uniform dangling row and a row concentrated on its out-neighbors, however small the weight involved. In the worked example of Section 2.13, adding an edge $D \to A$ of weight 10^{-12} moves $\mathbf{r}$ by 0.37 in the $\|\cdot\|_1$ norm and raises r_A from 0.1375 to 0.3202.

The damping sweep is covered by continuity: it keeps P fixed and changes d smoothly, and for $d < 1$ the PageRank vector varies continuously with d. The value-slope check is covered only in part. Since $V(z) > 0$ for every $z > 0$, a change of the slope b adds or removes no edge in exact

arithmetic, but in double precision an edge that carries only tiny amounts can round to zero weight, and a tenfold change of b is too large for the bound above to say much, so the stability of that check rests on the data (Table 4.7). Empirically, the family means of EndorseRank and AWP differ by less than 0.01 across $d \in \{0.75, 0.85, 0.95\}$ (Table 4.6).

8.3.3 Sparsity and per-iteration cost

With $m = |E|$ nonzero edges, the sparse product $P^{\top}\mathbf{x}$ costs $O(m)$ arithmetic operations, and building the sparse operator from the edge list is also $O(m)$, so power iteration over I iterations is $O(m+mI)$. On the matched cohort the transfer graph has 2.23 times as many edges as the allowance graph, so even at identical I the transfer method would run about 2.23 times as long. Both scores run the same solver. In the measured calls AWP needs more iterations than EndorseRank, yet the time ratio stays slightly below the edge ratio (Section 4.2), so on these graphs the transfer method spends less time per edge and iteration.

8.3.4 Properties of Spearman's ρ

Spearman's ρ lies in $[-1, 1]$. A value of $\rho = 1$ means perfect monotone agreement, and $\rho = -1$ perfect reverse agreement. The evaluation pipeline computes ρ and τ_b with standard tie-aware implementations. When n is large, ρ and τ usually agree in sign, but they differ in magnitude. Across the fourteen pairs of EndorseRank or AWP with a proxy (Tables 8.7 to 8.9), every $|\tau|$ exceeds 0.07, the ratio ρ/τ lies between about 1.2 and 1.8, and $|\rho|$ is always the larger of the two; no pair has coefficients of opposite sign.

8.3.5 Properties of Kendall's τ

Among the scores of this thesis ties are rare: on the matched cohort no score ties more than 3.2% of the contract pairs, and Section 4.4 gives the shares of EndorseRank and AWP. The denominator of τ_b in Equation (2.6) differs from n_0 only through ties, so for EndorseRank against AWP it differs from n_0 by less than two percent, and τ_a, computed from the pair counts of Section 4.4, lies within

two percent of the reported τ_b.

Ties in a proxy cap τ_b as well: the integer proxies, such as the in-degree and the in-approve degree, tie many pairs, and the largest τ_b that any score can reach against such a proxy falls with the share of tied pairs. Because the scores tie few pairs, this cap differs little between the scores compared on one proxy, so a paired contrast on one proxy is not distorted by it; a contrast between two proxies can be, to the extent that their ties differ.

8.4 Data Description

This appendix describes the extracted data behind Chapter 4: the volume of the ego extraction and of the two graphs, the five tokens with their prices and allowance amounts, and the activity of GMX V2 contract accounts by quarter.

8.4.1 Extracted events and graphs

The ego extraction of the observation window was made for the plan's cohort of $27{,}844$ contracts. It holds $389{,}866{,}882$ approval logs, zero-value ones included, and $1{,}538{,}404{,}067$ transfer logs of non-zero value of every ERC-20 token, each with a contract of that cohort at one end. The revised analysis keeps the approvals and transfers that are of the five tokens and have a contract of the revised cohort at one end (Section 4.1 gives their number); of the July–September logs of W1 it keeps $12{,}209{,}894$ approvals and $60{,}182{,}192$ transfers. At T_{obs} the pair tables built from the kept events hold $11{,}219{,}116$ allowance pairs and $25{,}018{,}892$ transfer pairs, and at t_1, $10{,}650{,}258$ and $23{,}680{,}692$. The $9{,}002$ and $17{,}676$ pairs at T_{obs} whose value weight is zero in double precision, and the addresses that only they touch (Section 4.1), are not part of the solver graphs of EndorseRank and AWP (Table 4.1).

8.4.2 Tokens, prices and allowance amounts

Section 3.5.1 gives the rule by which the five tokens were chosen. They hold 75.75% of the rows of all tokens in the ego tables of the plan's cohort (transfers up to T_{obs} and positive latest allowances

at T_{obs}). Table 8.4 gives for each of them the address, decimals and feed, the rows held and kept, the US-dollar volume of the kept transfers and the data behind the daily prices; its caption counts the rows left out, those of the other tokens and those of the five tokens in which no contract of the revised cohort takes part.

Table 8.4: The five tokens of the revised analysis, with their Chainlink feed (market-risk category), the rows of the observation window that the ego tables of the plan's cohort hold for them (approvals and transfers) and the rows that the revised analysis keeps, the US-dollar volume of the kept transfers, the days of the price series filled from the second DefiLlama key, filled from the previous day and replaced as one-day errors (of 1,096 days), and the median and largest absolute difference, in percent, between the DefiLlama price and the Chainlink feed read on the chain on 38 dates. The ego tables of the plan's cohort hold 389,866,882 approvals and 1,538,404,067 transfers in the observation window. Of these, 164,202,859 approvals and 364,468,917 transfers belong to other tokens (of the 142,958 other tokens in the ego tables of the observation window and W1) and are left out; 709,821 approvals and 24,766,612 transfers of the five tokens are left out because neither end point is a contract of the revised cohort; and 0 kept transfers precede their token's first price. At T_{obs}, 5,763,757 of the 12,590,613 positive latest allowances are unlimited and 6,776,289 reach the cap $U = 20{,}000$ dollars of the allowance layer of C-PR.

Token	Address	Dec.	Chainlink feed	Rows, plan	Rows, kept	Transfer USD	2nd key	Filled	Errors	CL diff. (%)
WETH	`0x82af…bab1`	18	ETH / USD (low)	586,587,366	569,011,073	8.4×10^{11}	319	0	0	0.06 / 0.72
USDC	`0xaf88…5831`	6	USDC / USD (low)	400,716,805	396,280,336	9.9×10^{11}	0	0	0	0.01 / 0.26
USDT0	`0xfd08…cbb9`	6	USDT / USD (low)	233,074,004	231,428,275	2.7×10^{11}	481	0	0	0.03 / 0.27
ARB	`0x912c…6548`	18	ARB / USD (low)	90,752,065	89,229,800	8.5×10^{10}	0	0	0	0.08 / 0.32
WBTC	`0x2f2a…5b0f`	8	WBTC / USD (low)	88,468,933	88,173,256	2.2×10^{11}	308	0	0	0.06 / 0.24

DefiLlama's series under the Arbitrum One address begins in August 2024 for WETH and WBTC and in January 2025 for USDT0, so their earlier days take the price of the same asset under its CoinGecko identifier; the two series differ by less than 0.1% in median on the days both cover. No day of the five tokens is filled from the previous day or replaced as a data error, and no kept transfer precedes its token's first price. Against the Chainlink feeds read on the chain on 38 dates per token, the DefiLlama prices differ by a median of 0.01% to 0.08% per token and by at most 0.72%, for WETH.

Unlimited allowances are common. Of the $12{,}590{,}613$ positive latest allowances at T_{obs}, $5{,}763{,}757$ (45.8%) are of 1.15×10^{77} base units or more, close to $2^{256} - 1$, and $6{,}776{,}289$ (53.8%), the unlimited ones included, are worth at least the cap U. The $12{,}590{,}613$ allowances belong to $6{,}208{,}769$ owners,

of whom $3{,}294{,}370$ hold at least one unlimited allowance and $797{,}852$ hold both an unlimited and a finite one. In the registered analysis, which kept every token in raw base units, 99.52% of the ego transfers and 99.78% of the $28{,}069{,}418$ latest allowances at T_{obs} were of ten base units or more, where V with $b = 1$ is within 10^{-4} of one, and $1{,}532{,}301$ of the $13{,}029{,}306$ owners with a positive latest allowance held both an unlimited and a smaller allowance.

8.4.3 GMX V2 contract traders by quarter

The trader cohorts are GMX V2 accounts that hold contract code (Section 4.1.1). Table 8.5 counts their position closes, the `PositionDecrease` events of the protocol's event emitter (GMX, n.d.), by calendar quarter, together with the liquidations among them and the number of contract accounts that closed at least one and at least three positions in the quarter.

Table 8.5: GMX V2 closes of contract accounts by quarter. October 2023 to June 2026: closes by the $7{,}633$ contracts among the $40{,}307$ GMX V2 accounts with at least three closes in the observation window (Table 8.1). July to September 2026 (the label window of W1): closes by the contract accounts of the observation window and of the W1 run. Liquidation: a close with the liquidation order type or emitted through the liquidation handler. Source: the decoded closes in `data/2-processed-tables-and-evaluations/gmx/`.

Quarter	Closes	Liquidations	Accounts with a close	Accounts with ≥ 3 closes
2023 Q4	1,119	104	180	105
2024 Q1	7,956	692	1,247	961
2024 Q2	11,509	589	2,120	1,165
2024 Q3	7,947	456	1,807	724
2024 Q4	7,343	904	1,321	698
2025 Q1	6,929	498	497	273
2025 Q2	12,080	379	368	250
2025 Q3	28,753	897	1,750	1,567
2025 Q4	40,240	1,596	1,915	1,744
2026 Q1	6,026	233	269	171
2026 Q2	3,783	125	169	127
2026 Q3 (W1)	6,805	182	165	131
October 2023 – June 2026	133,685	6,473		

Contract accounts closed the most positions in the second half of 2025 and far fewer in 2026. A

trader cohort further requires the account to be a node of the graph at the freeze: of the 127 contract accounts with at least three closes in April–June 2026, 88 form the trader cohort of W0, and of the 131 in July–September, 70 form that of W1.

8.5 EndorseRank and AWP Side by Side

Chapter 4 reports every score of the thesis together and reads each PageRank against the raw degree it smooths. This appendix collects the direct comparisons between the two single-layer scores, EndorseRank and AWP: their family means and paired differences in the same window, their per-proxy coefficients, and their paired differences out of window. Apart from the trader grid at its end, none of these comparisons was fixed in advance, and none carries a claim of the thesis. The two scores walk on different edges and are built for different constructs, so on each check the score ahead is, almost every time, the one whose edges the check counts. Outcomes built from neither edge type are the exception, since neither score was built for them, and are reported in Section 4.6.3. Their coverage differs as well: no allowance edge touches 523 of the matched contracts, and no transfer edge touches 2,670. Out of window AWP ranks later new senders above the transfer in-degree, while EndorseRank ranks later new approval pairs below the in-approve degree (Section 4.6), so a lead of EndorseRank over AWP does not show that propagation over approvals adds anything to the count of approving owners.

8.5.1 Same window

Table 4.3 gives the family means of the two scores with their intervals, and the last row of Table 4.5 their agreement. The two patterns cross: AWP tracks the transfer family most closely and the allowance family least, while EndorseRank tracks the allowance family most closely. Table 8.6 tests five paired differences on the same contract resamples (Section 3.10), numbered i to v in the order of its rows.

All five intervals lie above zero. AWP's transfer alignment is higher than EndorseRank's (contrast ii,

Table 8.6: Paired bootstrap tests of differences between Kendall τ coefficients of EndorseRank and AWP in the same window (matched cohort, $n = 15{,}107$ contracts). Family entries use the family-mean τ; the same contract resamples are used for a and b, so the interval is on the paired difference. An interval excluding 0 indicates a difference not attributable to sampling variation at the 5% level. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Contrast (a minus b)	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
EndorseRank allowance minus EndorseRank transfer	0.585	0.158	+0.427	[0.416, 0.440]	1.000
AWP transfer minus EndorseRank transfer	0.572	0.158	+0.414	[0.401, 0.427]	1.000
EndorseRank allowance minus AWP allowance	0.585	0.193	+0.392	[0.380, 0.405]	1.000
AWP sybil-stability minus EndorseRank sybil-stability	0.395	0.007	+0.388	[0.376, 0.401]	1.000
EndorseRank in-approve degree minus EndorseRank in-degree	0.655	0.182	+0.473	[0.461, 0.486]	1.000

$+0.414$, $[0.401, 0.427]$), EndorseRank's allowance alignment is higher than AWP's (contrast iii, $+0.392$, $[0.380, 0.405]$), and AWP leads EndorseRank on the Sybil-stability family (contrast iv, $+0.388$, $[0.376, 0.401]$). The remaining two compare EndorseRank with itself: its allowance alignment is higher than its transfer alignment (contrast i, $+0.427$, $[0.416, 0.440]$), and its in-approve-degree alignment is higher than its in-degree alignment (contrast v, $+0.473$, $[0.461, 0.486]$). EndorseRank ties only 0.17% of the contract pairs, so its own ties cap none of these coefficients, and the ties of the proxies do not explain gaps of this size (Appendix 8.3.5). With the contracts outside the allowance graph kept as isolated nodes instead of being scored zero, none of the family means changes (Table 4.5).

The opposite directions of contrasts ii and iii follow from the constructs. Each score is ahead on the family built from its own edges, and the main proxies of each family include the in-degree that the corresponding PageRank smooths, so the agreement is partly built in and checks the intended construct of each score. Neither difference says which score is the better measure of reputation. On every scaling stage of Table 4.8 the order of the two scores within each family is the same as on the full cohort.

8.5.2 Proxy detail

Each family mean condenses several per-proxy coefficients. Tables 8.7 to 8.9 report them one proxy at a time for EndorseRank and AWP, as Spearman ρ, Kendall τ and the 95% interval of τ;

the transfer proxies are the ones used in the evaluation of AWP by its authors (T. Do et al., 2023).

8.5.2.1 Transfer family

Table 8.7 gives the coefficients of the two scores with the in-degree and the in-value.

Table 8.7: Domain alignment of EndorseRank and AWP with the transfer proxies (in-degree and in-value of each contract), same window (matched cohort, $n = 15{,}107$ contracts). Kendall τ and Spearman ρ between scores and proxy rankings. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
EndorseRank			
In-degree	0.261	0.182	[0.169, 0.194]
In-value (sum inbound)	0.205	0.134	[0.123, 0.145]
AWP			
In-degree	0.744	0.578	[0.569, 0.587]
In-value (sum inbound)	0.741	0.566	[0.558, 0.575]

AWP agrees with the in-value almost as closely as with the in-degree (Table 8.7), for the reason given in Section 4.3. EndorseRank's transfer coefficients have intervals that overlap neither zero nor AWP's intervals. They mainly reflect that contracts that many owners approve receive transfers from more senders than the rest of the cohort.

8.5.2.2 Allowance family

Table 8.8 gives the coefficients of the two scores with the in-approve degree and the in-approve value.

EndorseRank follows the in-approve degree more closely than the in-approve value, for the reason given in Section 4.3. AWP's coefficients lie above zero: among the cohort's contracts, those that many owners approve also tend to be central in the transfer graph. The comparison is partly built in, since EndorseRank is computed from the same approvals.

Table 8.8: Domain alignment of EndorseRank and AWP with the allowance proxies (in-approve degree and value of each spender contract), same window (matched cohort, $n = 15{,}107$ contracts). Kendall τ and Spearman ρ between scores and proxy rankings. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
EndorseRank			
In-approve degree	0.811	0.655	[0.648, 0.662]
In-approve value (sum)	0.681	0.515	[0.506, 0.524]
AWP			
In-approve degree	0.286	0.206	[0.193, 0.217]
In-approve value (sum)	0.260	0.180	[0.169, 0.191]

8.5.2.3 Sybil-adjusted stability family

Table 8.9 gives the coefficients of the two scores with the inbound counterparty ratio, the transfer tenure and the active months.

Table 8.9: Domain alignment of EndorseRank and AWP with the Sybil-adjusted stability proxies from transfer activity, same window (matched cohort, $n = 15{,}107$ contracts). Kendall τ and Spearman ρ between scores and proxy rankings. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
EndorseRank			
Inbound counterparty ratio	-0.231	-0.156	[-0.166, -0.147]
Transfer tenure (days)	0.124	0.080	[0.069, 0.091]
Active months	0.146	0.098	[0.087, 0.109]
AWP			
Inbound counterparty ratio	0.234	0.127	[0.114, 0.141]
Transfer tenure (days)	0.683	0.508	[0.499, 0.517]
Active months	0.704	0.549	[0.540, 0.558]

AWP is ahead of EndorseRank on tenure and active months, as one would expect from a transfer score that rewards activity spread over the window; the paired difference on the family mean excludes zero (Table 8.6, contrast iv). On the inbound counterparty ratio the two coefficients have opposite signs, negative for EndorseRank and positive for AWP. The ratio falls when the same senders

return, so contracts that many owners approve are, on this measure, contracts whose senders return. AWP's positive coefficient comes from coverage: the $2{,}670$ contracts outside the transfer graph have ratio zero and AWP score zero, so they fill the bottom of both orderings. The family describes how much longitudinal transfer information each score carries.

8.5.3 Out of window

Table 8.10 collects the paired differences between the two scores in both holdout windows. In every row a is EndorseRank and b AWP.

Table 8.10: Paired bootstrap differences between EndorseRank and AWP out of window (W0 spenders $n = 17{,}050$; W0 traders $n = 88$; W1 spenders $n = 18{,}152$; W1 traders $n = 70$). None of them enters rule A or rule B: the spender rows were computed after the labels were known, and the trader rows are the EndorseRank-minus-AWP pair of the comparison on an outcome built from neither edge type (Tables 4.18 and 8.11). The W1 spender rows were computed from the saved W1 scores after the W1 evaluation, with the resamples of Table 4.14. Within each block a and b share the contract resamples (400 for the spenders and 2,000 for the traders; seed 42).

Contrast (a minus b)	Label	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
W0, spenders ($n = 17{,}050$)						
EndorseRank minus AWP	new approval pairs	0.231	0.196	+0.035	[0.019, 0.051]	1.000
EndorseRank minus AWP	new transfer senders	0.200	0.346	-0.146	[-0.156, -0.136]	0.000
W0, traders with at least three closes ($n = 88$)						
EndorseRank minus AWP	liquidation-free close rate	0.224	0.240	-0.017	[-0.156, 0.128]	0.394
W1, spenders ($n = 18{,}152$)						
EndorseRank minus AWP	new approval pairs	0.234	0.169	+0.065	[0.051, 0.081]	1.000
EndorseRank minus AWP	new transfer senders	0.194	0.330	-0.136	[-0.146, -0.127]	0.000
W1, traders with at least three closes ($n = 70$)						
EndorseRank minus AWP	liquidation-free close rate	0.297	0.329	-0.032	[-0.133, 0.066]	0.279

On the spenders of W0, EndorseRank ranks new approval pairs above AWP, a paired difference of $+0.035$ (interval $[0.019, 0.051]$). On new transfer senders the order reverses, with a difference of -0.146 ($[-0.156, -0.136]$). Every label falls after all scored events, so on this cohort each edge type carries information about its own construct that the other lacks. Coverage accounts for part of the first gap. AWP scores zero for the $2{,}891$ spenders without a transfer edge before the freeze, and

these gained new approval pairs more often than the rest. On the $14{,}094$ spenders that had received a transfer from another address, AWP is ahead on both labels (Table 4.12), without a paired test.
In July–September, with the contrasts computed from the W1 scores after all labels were known, the pattern repeats: EndorseRank leads on new approval pairs ($+0.065$, $[0.051, 0.081]$) and AWP on new senders (-0.136, $[-0.146, -0.127]$).
On the W0 traders (Table 4.13), only AWP's intervals lie above zero on the number of profitable closes and on the realized gain, two labels that grow with the number of closes. On the two shares, which do not, both scores are negative, with intervals below zero on the non-loss share only. Neither score ranks these contract traders by the share of successful closes.

8.5.4 All trader labels in both windows

Table 8.11 sets the allowance side against the transfer side on all six trader labels in both windows. Its pairs were registered with W1 before the study extracted the logs of July–September (Section 3.7.3); here they are computed on the revised scores, after the registered results were known, so the table is a re-analysis (Appendix 8.6). P and Q on the liquidation-free close rate enter rules R1 and R2, and P on the two shares enters rule R3 (Claim C6, Section 4.6.3). In W1 no pair separates the two sides on either liquidation label. In W0 the counts (P) favor the allowance side on both liquidation labels, while the three PageRank pairs lie below zero with intervals that include zero. On the four profit labels of W1 every pair favors the transfer side: the intervals of Q and of the EndorseRank pair lie below zero on all four, those of P on the two shares, and those of the C-PR layers on realized gain and on both shares. In W0 every interval on the profit labels includes zero except that of the C-PR layers on realized gain.

8.6 The Registered Analysis

The registered analysis is the one that the analysis plan and the registration of W1 fixed before the study extracted the data it judges (Appendix 8.1.1): amounts in raw token base units, every ERC-

Table 8.11: Allowance-side minus transfer-side scores, $\Delta\tau$ with 95% paired bootstrap interval, on all six trader labels of the window W1 ($n = 70$ GMX V2 contract traders) and of W0 ($n = 88$). P: in-approve degree minus transfer in-degree. Q: EndorseRank with activity restarts minus AWP (same edge weights and restart rule). C-PR: $\lambda = 1$ minus $\lambda = 0$ (same USD build). ER: EndorseRank minus AWP. Profitable closes and realized gain grow with the number of closes; the other four labels do not. CI = 95% percentile bootstrap over contracts (2,000 paired resamples, seed 42).

Label	P (counts)	Q (PageRanks)	C-PR layers	ER minus AWP
Window W1, July–September 2026 ($n = 70$ traders)				
Liquidation-free close rate	0.084 [-0.013, 0.176]	-0.041 [-0.135, 0.050]	-0.049 [-0.169, 0.060]	-0.032 [-0.133, 0.066]
No liquidation	0.078 [-0.012, 0.166]	-0.039 [-0.138, 0.062]	-0.051 [-0.170, 0.064]	-0.029 [-0.136, 0.080]
Profitable closes	-0.143 [-0.318, 0.013]	-0.231 [-0.429, -0.027]	-0.190 [-0.411, 0.037]	-0.214 [-0.423, -0.012]
Realized gain	-0.135 [-0.308, 0.025]	-0.350 [-0.536, -0.157]	-0.404 [-0.624, -0.182]	-0.346 [-0.547, -0.144]
Profitable share	-0.196 [-0.364, -0.046]	-0.336 [-0.523, -0.141]	-0.344 [-0.551, -0.120]	-0.316 [-0.517, -0.112]
Non-loss share	-0.252 [-0.391, -0.120]	-0.350 [-0.526, -0.175]	-0.324 [-0.531, -0.109]	-0.346 [-0.526, -0.161]
Window W0, April–June 2026 ($n = 88$ traders)				
Liquidation-free close rate	0.182 [0.052, 0.314]	-0.014 [-0.147, 0.130]	-0.041 [-0.187, 0.105]	-0.017 [-0.156, 0.128]
No liquidation	0.197 [0.059, 0.333]	-0.010 [-0.152, 0.146]	-0.051 [-0.198, 0.102]	-0.012 [-0.154, 0.137]
Profitable closes	0.075 [-0.080, 0.222]	-0.016 [-0.150, 0.118]	-0.067 [-0.222, 0.087]	-0.005 [-0.163, 0.155]
Realized gain	0.017 [-0.150, 0.176]	-0.107 [-0.252, 0.044]	-0.248 [-0.433, -0.067]	-0.118 [-0.286, 0.046]
Profitable share	0.045 [-0.113, 0.184]	0.046 [-0.071, 0.172]	0.027 [-0.112, 0.181]	0.069 [-0.060, 0.208]
Non-loss share	-0.080 [-0.228, 0.058]	0.055 [-0.070, 0.192]	0.037 [-0.111, 0.189]	0.076 [-0.063, 0.223]

20 token, the value transform with $b = 1$, and the plan's cohort of $27{,}844$ contracts. The amounts were revised after its results were known (Section 3.5.1), and the chapters report the revised analysis. Table 8.12 sets the decision contrasts and verdicts of the registered analysis beside the same contrasts on the revised scores; its last four rows, degree contrasts computed after the labels were known, carry no verdict. Both columns reach the same verdicts, but only those of the registered column are outcomes of tests fixed before the study extracted the data they judge. The complete registered results, with every coefficient, interval and verdict, are published unchanged in `data/3-published-results-for-thesis/` with their digest, `results_digest.md`.

8.7 Ethics Clearance Certificate

8.7.1 Certificate

This study uses public blockchain data only and involves no human participants (Section 3.16). The ethics clearance certificate issued by the Unisa School of Computing / College of Agriculture and Environmental Sciences research ethics review committee is reproduced below.

Ethics clearance certificate

Reference: [Unisa SoC/CAES ethics clearance ref: TBD]

Date of issue: [TBD]

(Scanned certificate to be inserted here.)

Table 8.12: Decision contrasts and verdicts of the registered analysis (amounts in raw base units, every ERC-20 token, $b = 1$), exactly as obtained under the analysis plan and the W1 registration, beside the same contrasts on the revised scores (amounts in USD, listed tokens only). The revision was decided after the registered results were known; the revised column applies the registered rules and thresholds to the revised scores and is not a registered test. The last block holds degree contrasts that no rule of either analysis decides; they are reported in both. $\Delta\tau$ with 95% paired bootstrap interval (400 and 2,000 resamples, seed 42); the rules and their thresholds are in Sections 3.7 and 3.7.2.

Quantity	Registered analysis	Revised analysis
Matched cohort (contracts)	27,844	15,107
W0 spender cohort	33,101	17,050
W1 spender cohort	34,587	18,152
W1 traders	70	70
EndorseRank vs. AWP, same window (τ)	0.217 [0.209, 0.225]	0.174 [0.163, 0.185]
A, W0: C-PR minus C-PR ($\lambda = 1$), new approval pairs	$+0.008$ [0.003, 0.013]	$+0.029$ [0.022, 0.036]
A, W0: C-PR minus C-PR ($\lambda = 0$), new transfer senders	-0.030 [-0.034, -0.026]	-0.034 [-0.040, -0.029]
Rule A, primary part	not met	not met
Rule A, secondary part	not met	not met
B, F1: C-PR minus C-PR ($\lambda = 0$), new approval pairs	$+0.101$ [0.092, 0.111]	$+0.110$ [0.098, 0.122]
B, F2: C-PR minus C-PR ($\lambda = 0$), new transfer senders	-0.025 [-0.029, -0.021]	-0.028 [-0.033, -0.023]
B, F3: C-PR minus C-PR ($\lambda = 0$), liquidation-free close rate	-0.044 [-0.100, -0.002]	$+0.000$ [-0.014, 0.010]
Rule B (F1, F2 and F3)	not met	not met
Rule B, sensitivity 1 (AWP as comparator)	not met	not met
Rule B, sensitivity 2 (isolated nodes)	not met	not met
F6 (rule A, primary part, on W1)	fails	fails
P, W1: in-approve degree minus transfer in-degree	$+0.063$ [-0.026, 0.148]	$+0.084$ [-0.013, 0.176]
Q, W1: EndorseRank (activity restarts) minus AWP	$+0.051$ [-0.085, 0.213]	-0.041 [-0.135, 0.050]
R1 (count-level lead)	not met	not met
R2 (PageRank-level lead)	not met	not met
R3 (specificity)	applies	applies
W0: EndorseRank minus in-approve degree, new pairs	-0.011 [-0.016, -0.006]	-0.019 [-0.026, -0.011]
W0: AWP minus transfer in-degree, new senders	$+0.038$ [0.035, 0.042]	$+0.043$ [0.038, 0.048]
W1: EndorseRank minus in-approve degree, new pairs	-0.029 [-0.032, -0.025]	-0.034 [-0.038, -0.029]
W1: AWP minus transfer in-degree, new senders	$+0.020$ [0.016, 0.023]	$+0.022$ [0.017, 0.027]